

Foundation (Jalapeno)

- Q1.** What is the standard form of a quadratic function?
 (A) $ax^2 + bx + c$
 (B) $ax + bx^2 + c$
 (C) $ax^2 - bx + c$
 (D) $ax^3 + bx^2 + c$
 (E) $ax^2 + b$
- Q2.** A stock's price is modeled by the function $f(x) = 2x^2 - 5x + 3$. What type of function is this?
 (A) Linear
 (B) Quadratic
 (C) Cubic
 (D) Exponential
 (E) Trigonometric
- Q3.** The function $f(x) = x^2 - 4x + 4$ can be written in the form $f(x) = a(x - h)^2 + k$. What is the value of a ?
 (A) 1
 (B) 2
 (C) 3
 (D) 4
 (E) 5
- Q4.** A company's profit is modeled by the function $f(x) = -x^2 + 10x - 20$. What is the vertex of the parabola?
 (A) (0, -20)
 (B) (5, 5)
 (C) (10, 0)
 (D) (5, 15)
 (E) (0, 0)
- Q5.** The function $f(x) = 3x^2 + 2x - 1$ has a key feature of
 (A) A maximum value
 (B) A minimum value
 (C) A zero value
 (D) A vertical asymptote
 (E) A horizontal asymptote
- Q6.** A quadratic function $f(x) = ax^2 + bx + c$ has $a = 2$, $b = -5$, and $c = 1$. What is the equation of the function?
 (A) $f(x) = 2x^2 + 5x + 1$
 (B) $f(x) = 2x^2 - 5x + 1$
 (C) $f(x) = 2x^2 + 5x - 1$
 (D) $f(x) = x^2 - 5x + 1$
 (E) $f(x) = x^2 + 5x + 1$
- Q7.** A savings account earns interest according to the function $f(x) = 0.05x^2 + 10x + 100$. What type of function models the interest?
 (A) Linear
 (B) Quadratic
 (C) Exponential
 (D) Logarithmic
 (E) Trigonometric
- Q8.** The graph of $f(x) = x^2 - 2x - 3$ opens
 (A) Upward
 (B) Downward
 (C) Leftward
 (D) Rightward
 (E) Neither upward nor downward

Open Ended Questions (FRQ)

- Q9.** Describe the key features of the quadratic function $f(x) = -2x^2 + 4x + 1$. Provide examples of how the function could be used in a business context.
- Q10.** Write a quadratic function in standard form that models a company's profit based on the number of products sold. Assume the profit is 100 when 10 products are sold and 150 when 15 products are sold. Show your work and explain your reasoning.

Chef's Tips

- Emphasize the importance of identifying the standard form of a quadratic function
- Provide examples of how quadratic functions can be used in real-world business contexts
- Encourage students to analyze and describe the key features of quadratic functions